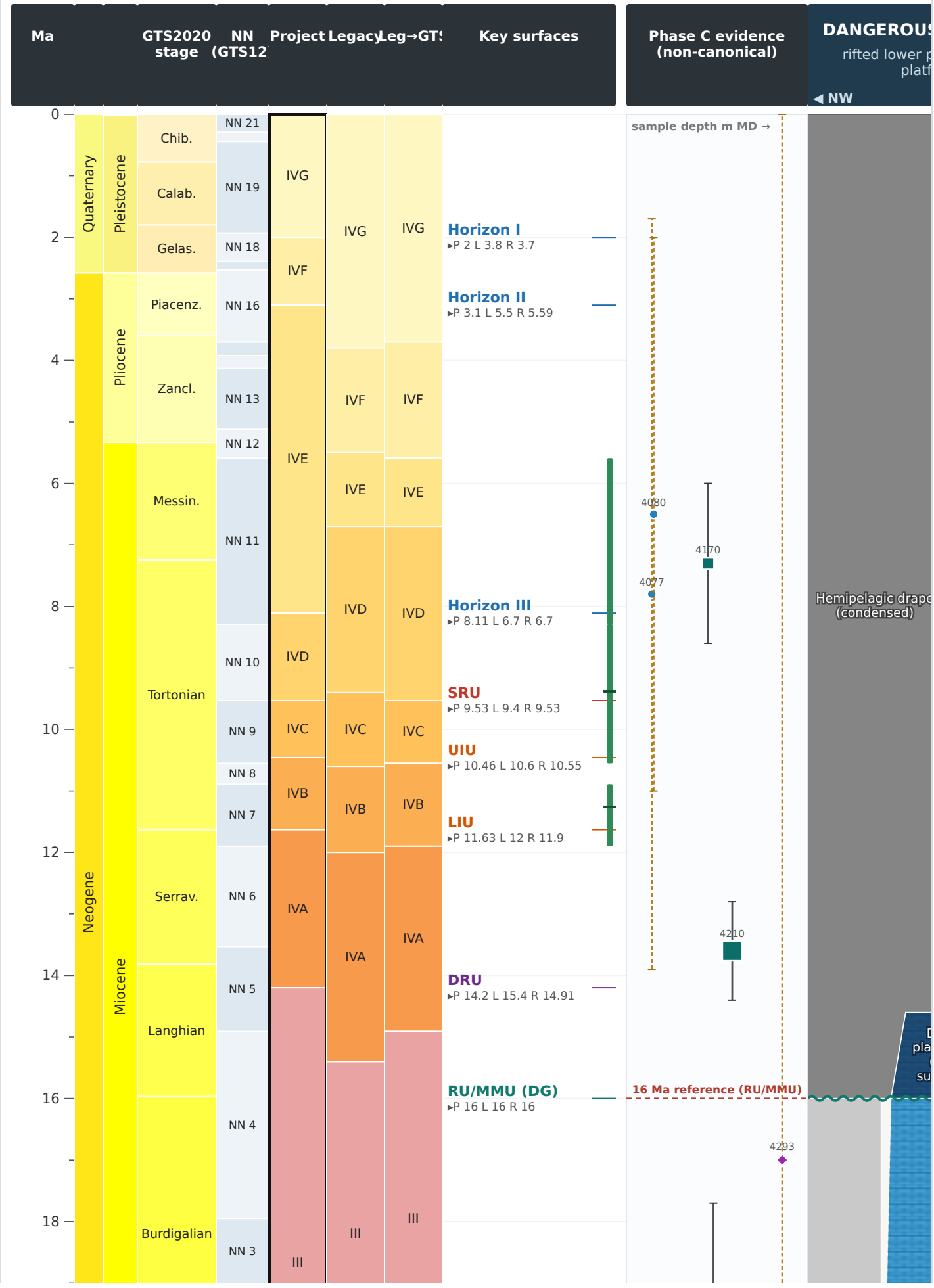
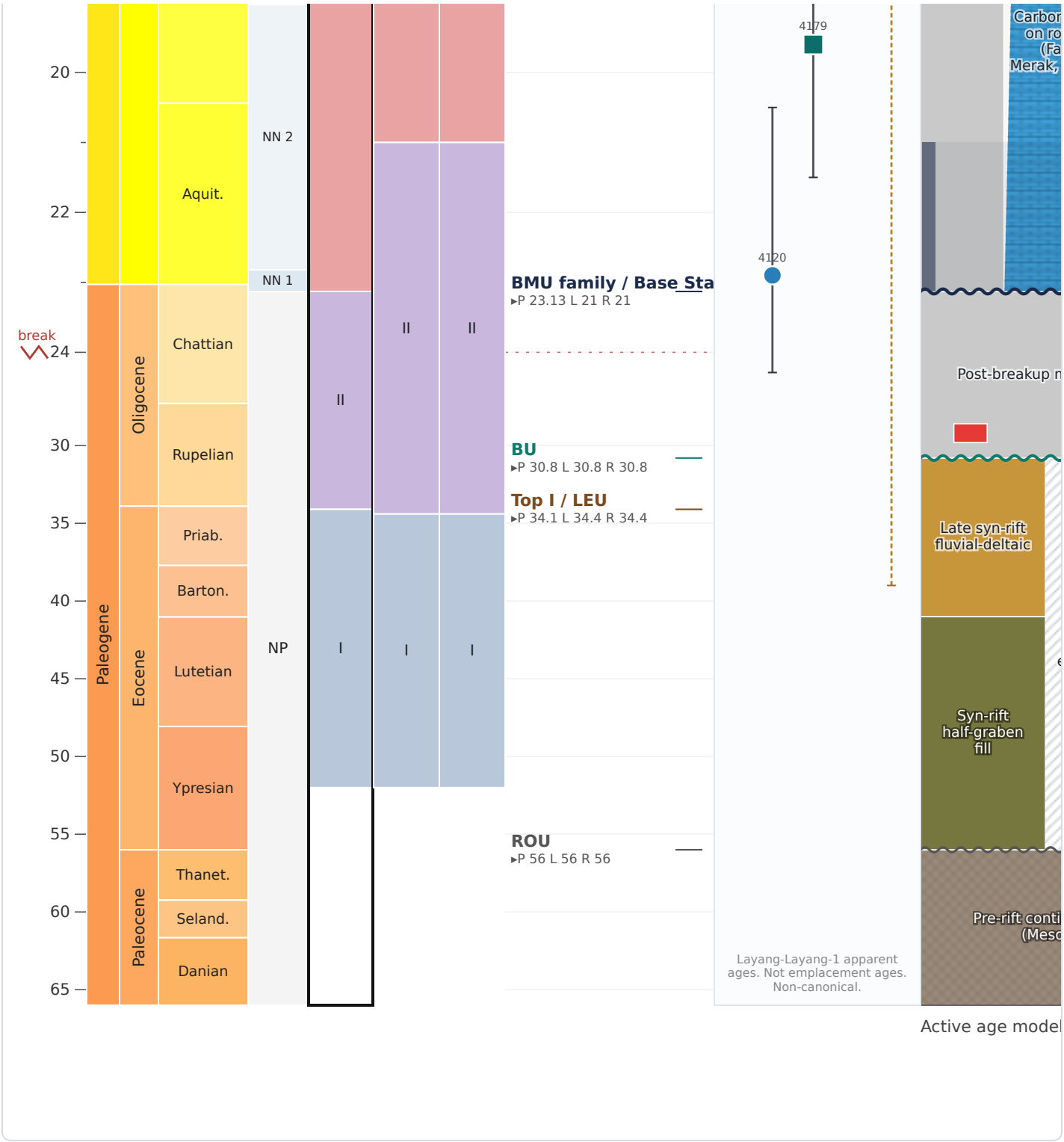
Kinabalu Basin: Chronostratigraphy & Tectonostratigraphy

NW–SE domain Wheeler chart, 0–66 Ma · v2.2-CANDIDATE · 2026-10-12 · rendered from kinabalu_chronostrat_v2.json + kinabalu_chronostrat_v2_2_ext.json

v2.2 CANDIDATE - WORKING MODEL, NOT SEALED **PETRONAS INTERNAL** **Build validation: 0 issues**

Top-of-stage ages from Kinabalu_Basin_Unified_Stratigraphic_Chart_GEOX.xlsx / Phase I marker standard (GTS2012-type nannofossil calibration).


Ma
GTS2020 stage
NN (GTS12
Project
Legacy
Leg→GTS
Key surfaces
Phase C evidence (non-canonical)
DANGEROUS
rifted lower p
platf
◀ NW
sample depth m MD →
Quaternary
Pleistocene
Pliocene
Miocene
Neogene
Chib.
Calab.
Gelas.
Piacenz.
Zancl.
Messin.
Tortonian
Serrav.
Langhian
Burdigalian
NN 21
NN 19
NN 18
NN 16
NN 13
NN 12
NN 11
NN 10
NN 9
NN 8
NN 7
NN 6
NN 5
NN 4
NN 3
IVG
IVF
IVE
IVD
IVC
IVB
IVA
III
Horizon I
▸P 2 L 3.8 R 3.7
Horizon II
▸P 3.1 L 5.5 R 5.59
Horizon III
▸P 8.11 L 6.7 R 6.7
SRU
▸P 9.53 L 9.4 R 9.53
UIU
▸P 10.46 L 10.6 R 10.55
LIU
▸P 11.63 L 12 R 11.9
DRU
▸P 14.2 L 15.4 R 14.91
RU/MMU (DG)
▸P 16 L 16 R 16
4080
4170
4077
4210
4293
16 Ma reference (RU/MMU)
Hemipelagic drape
(condensed)
0
2
4
6
8
10
12
14
16
18



Phase C — candidate evidence (non-canonical)

verdict: HOLD for causal interpretation **canonical: false** **confidence LOW-MED**

INTERPRET constrained by DATA Local apparent-age evidence exists; emplacement age unresolved.

Warning. Aggregate K-Ar measurements include mixed primary and secondary mineral domains. Apparent ages must not be treated as emplacement ages without petrographic separation and mineral-specific dating.

Depth order. Depth order does not produce a simple age progression: 7.8, 6.5, 22.9, 7.3, 19.6, 13.6, 17.0 Ma from 4077 to 4293 m MD.

Depth (m MD)	Method	Fraction	Apparent age (Ma)	2σ (Ma)	2σ range (Ma)	Radiogenic &sup4;&sup0;Ar %	Treatment
4077.2	K-Ar	>2 µm; disintegrated	7.8	±6.1	1.7–13.9	5%	LOW confidence; weak radiogenic fraction
4080.3	K-Ar	>2 µm; soft	6.5	±4.5	2–11	5.2%	LOW confidence; weak radiogenic fraction
4119.7	K-Ar	>2 µm; disintegrated	22.9	±2.4	20.5–25.3	38.8%	MED-HIGH as apparent age
4170	K-Ar	bulk; hard sample	7.3	±1.3	6–8.6	19.6%	MEDIUM; strongest local young apparent-age signal
4179	K-Ar	bulk; disintegrated	19.6	±1.9	17.7–21.5	44.2%	MED-HIGH as apparent age
4210	K-Ar	bulk; hard sample	13.6	±0.8	12.8–14.4	56.5%	MEDIUM apparent age; possible mixed or intermediate system
4292.5	U-Pb lower intercept	secondary carbonate	17	±22	0–39	n/a	VERY LOW; disturbed Pb-rich system

Three competing working models — none is preferred

Model A — Older magmatism plus younger drape / reworking

VIABLE MED

The old K-Ar population (19.6-22.9 Ma), volcaniclastic deposition and younger fossil signals without any new magma.

Falsifier: Replicated younger-than-16 Ma crystallisation ages from pristine primary minerals or chilled margins.

Model B — Genuine polyphase magmatism

PLAUSIBLE LOW-MED

An older Early Miocene system plus a discrete post-spreading pulse.

Requires:
- primary-mineral young ages
- a young cross-cutting intrusion
- a coherent younger geochemical population

Falsifier: Young ages occur only in secondary alteration phases and no young cross-cutting geometry exists.

Model C — Apparent youth without new magma

VIABLE MED

Alteration, partial isotopic resetting, hydrothermal circulation, redeposition or fault reactivation produce young signals.

Falsifier: A concordant young crystallisation age from a pristine primary mineral, fresh groundmass, zircon or baddeleyite confirmed to be primary.

Confidence ladder

▶ LOW-MED (current)

- local apparent-age signals younger than 16 Ma
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- ambiguous mineral domains and geometry

MED

EITHER concordant younger-than-16 Ma primary-mineral crystallisation ages from two screened in-situ samples
OR one well-tied intrusion demonstrably younger than a secure post-16 Ma horizon

MED-HIGH

BOTH independent tests above

Regional-analogue ledger (not local evidence)

Analogue	Value
Mount Kinabalu granite crystallisation Cottam et al.	7.85 ± 0.08 to 7.22 ± 0.07 Ma

(2010) J. Geol. Soc. 167 - Kinabalu granite U-Pb	
South China Sea spreading cessation Li et al. (2014) IODP Exp. 349 - SCS spreading history	preliminary 16-20 Ma range (IODP Expedition 349), pending refinement by post-cruise work

Rock physics, mechanics and seismic relative timing

Lanes required before any Phase C body can be characterised. Nothing is filled in that the project data do not contain.

AVO / interface elastic properties

UNKNOWN: actual interface-average elastic properties not encoded

No AVO class (e.g. IIp, IV) is attached to a rock body. AVO is an interface response, not a lithology or age classifier.

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Mechanical permissibility

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Present-day stress is a permissibility constraint only. It never dates a Miocene intrusion.

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Seismic relative timing

UNKNOWN: no candidate intrusive feature is encoded with mapped seismic geometry

Relationship rules and implied inequalities are strictly enforced before any magmatic event is promoted.

Relationship	Implied inequality
Dated sediment drapes the body	`age(body) > age(drape base) - body existed before the drape was deposited`

Intrusion cross-cuts a dated horizon	`age(intrusion) < age(crossed horizon) - intrusion is younger`
Forced fold deforms dated sediment	`age(deformation source) < age(youngest deformed package) - only if the causal link to the intrusion is demonstrated`
Multiple sill levels	`separate intrusive levels; NOT automatically separate ages`
Strata onlap the body	`age(body relief) >= age(oldest onlapping reflector) - write the inequality; onlap alone is not an age`

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QC flags: See well 'excluded' entries and contradiction C12 (Maligan-1 reversal, Rusa Timur-1 truncated label, Imbok-1 datum conflict, Kudat-1 '?Crocker' with NN9a).

What the evidence says

Findings computed from the encoded well data (GEOX build, local Python) and reconciled against the three age models. These are the points that change how the chart should be used.

SRU is about 9.4 Ma, not 8.6 Ma

9.41 · 9.36 · 9.38 Ma

Three wells place the SRU pick inside NN10, just above NN9 (Gajah Hitam-1, Tiga Papan-2, Maligan-1). '8.6 Ma' is the 1995-timescale age of base NN11, and NN10 overlies the pick at all three wells. Project 9.53 is ~0.15 Ma too old; legacy 9.4 fits best.

Both schemes name the same surfaces

top ≡ base

Project ages are top-of-stage, legacy ages base-of-stage: Top III = Base IVA = DRU, Top IVC = Base IVD = SRU. Copilot's charts mixed the two conventions (e.g. '9.53–11.63 = IVC–IVB').

Part of the DRU gap is calibration

41% explained

Legacy 15.4 Ma is the 1995 age of base NN5. Under GTS2012 it is 14.91 Ma, leaving a residual of 0.71 Ma against the project's 14.2. No DRU pick in the pen chart has NN control, so it remains on HOLD.

Upper Stage IV is a definitional split

1.7–2.4 Myr

Base IVF (3.1 vs ~5.5) and Base IVG (2.0 vs 3.8) differ far beyond calibration error. These are two different IVE–IVG definitions under the same names. Decide before Phase II mapping.

UIU and LIU support the project ages

UIU 9.53–10.55 Ma

Project 10.46 (UIU) and 11.63 (LIU) pass every well test. Legacy 10.6 and 12.0 fail at Glayzer-3 and Imbok-1.

Phase C is a candidate, not an event

LOW–MED · non-canonical

Layang-Layang-1 gives young apparent ages (6.5–7.8, 13.6 Ma) interleaved with old ones (19.6–22.9 Ma) and no depth progression. None is a primary-mineral crystallisation age. Models A, B and C all remain open (C15–C18).

Shoaling is diachronous; highs are composite

C8 · C13

Stage III is bathyal inboard. Coastal facies appear on highs right after DRU, but the outer shelf stays bathyal until SRU–IVE time. On inversion highs, 'SRU' and 'DRU' labels sit on amalgamated surfaces with NN5–NN8 or IVB–IVE missing.

Phase C — candidate evidence (non-canonical)

verdict: HOLD for causal interpretation **canonical: false** **confidence LOW-MED**

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MED-HIGH

BOTH independent tests above

Regional-analogue ledger (not local evidence)

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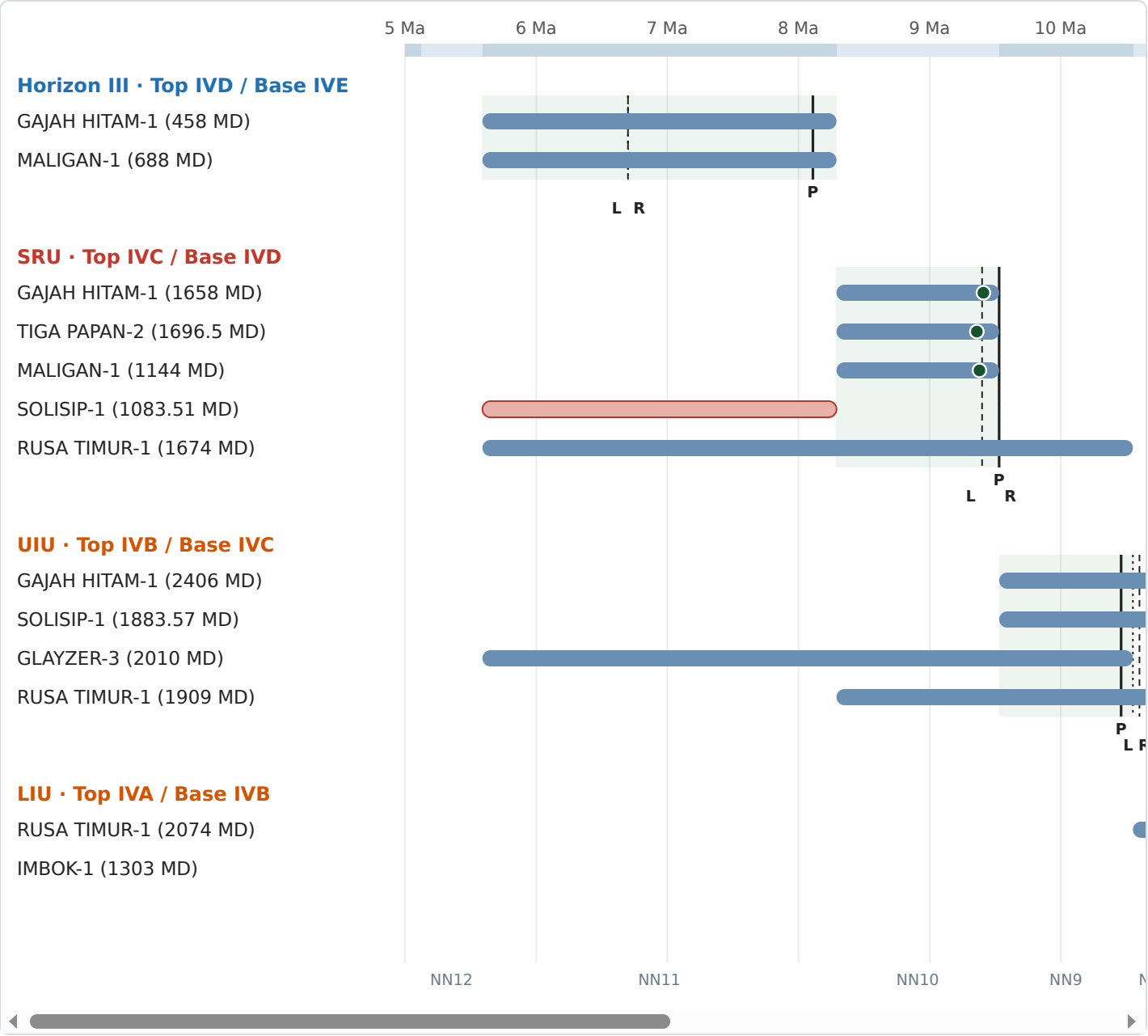
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Well witness: surface age windows (GTS2012)

Each bar shows the age window allowed by the nannofossil zones directly above and below a surface pick: the surface must be younger than the rock below it and older than the rock above it. Dots are constant-rate interpolations inside a fully bracketed zone (INT tier). Ticks show the three model ages; the shaded band is the joint window of the non-outlier wells.



Phase C — candidate evidence (non-canonical)

verdict: HOLD for causal interpretation | canonical: false | confidence LOW-MED

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Surface reconciliation

Surfaces are the stored objects; stages are the intervals between them. Project ages are top-of-stage ages and legacy ages are base-of-stage ages, but they name the same surfaces.

Surface	Definition	Project 2026	Legacy TB	Legacy → GTS2012	|P−L|	Gap removed by recal.	Well picks tested	Pass P / L / R	Witness window
H1	Top IVF / Base IVG	2	3.8	3.7	1.8	6%	0	-	
H2	Top IVE / Base IVF	3.1	5.5	5.59	2.4	-	0	-	
H3	Top IVD / Base IVE	8.11	6.7	6.7	1.41	-	2	2 / 2 / 2	5.59–8.2
SRU	Top IVC / Base IVD	9.53	9.4	9.53	0.13	100%	5	1 / 4 / 1	8.29–9.5
UIU	Top IVB / Base IVC	10.46	10.6	10.55	0.14	36%	4	4 / 3 / 3	9.53–10.5
LIU	Top IVA / Base IVB	11.63	12	11.9	0.37	27%	2	2 / 1 / 1	10.89–11.
DRU	Top III / Base IVA	14.2	15.4	14.91	1.2	41%	0	-	
BMU	Top II / Base III	23.13	21	21	2.13	-	0	-	

MMU	DG-domain surface; not the DRU	16	16	16	0	-	0	-
BU	onset SCS spreading	30.8	30.8	30.8	0	-	0	-
LEU	Top I / Base II	34.1	34.4	34.4	0.3	-	0	-
ROU	onset of SCS-margin stretching	56	56	56	0	-	0	-

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Contradiction register

Each HOLD has a discriminating test. Nothing here is averaged away.

ID	Topic	State	Models	Finding	Discriminating test
C1	DRU age	**HOLD-NARROWED**	P 14.2 | L 15.4 | R 14.91	The 15.4-15.5 Ma legacy age is the old-timescale age of base NN5 (BKSA95 15.6). Re-dated with GTS2012 it becomes 14.91, so ~0.5 Ma of the 1.2 Ma gap is a calibration artefact. The residual 0.71 Ma is real: the project places DRU inside NN5, not at its base.	No DRU pick in the pen chart has NN control (Deluar-1, Ketam-1, South Lokan-1 are EOD-only). Pull raw NN4/NN5 occurrences at these wells; if NN5 is present below the DRU pick, R (14.91) is falsified and P survives.
C2	SRU age (8.6 vs 9.4 vs 9.53)	**RESOLVED-PROVISIONAL: ~9.4 Ma (3 wells)**	P 9.53 | L 9.4 (alt 8.5) | R 9.53	At Gajah Hitam-1, Tiga Papan-2 and Maligan-1 the SRU pick lies inside NN10 (GTS2012 window 8.29-9.53 Ma). Constant-rate interpolation inside the bracketed NN10 span gives 9.41 / 9.36 / 9.38 Ma, so the legacy 9.4 Ma fits best. Project 9.53 is the base-NN10 approximation and sits on the window edge: it would need zero NN10 below the pick, but 8.5-110 m are observed. It is ~0.15 Ma too old, which does not matter for mapping. '8.6 Ma' is the BKSA95 age of base NN11. The hypothesis 'SRU = base NN11' (8.29 Ma GTS2012) is rejected because NN10 overlies the pick at all three wells. Solisip-1 is the outlier: NN11 lies below its B.S IVD pick (window 5.59-8.29), so either that pick is a younger surface or SRU is	Re-tie Solisip-1 B.S IVD on seismic; trace SRU terminations from Maligan-1 / Tiga Papan-2 to Solisip-1 along KL2. Add more NN10-bracketed SRU picks to test the 9.4 Ma estimate.

				diachronous toward N Sabah.	
C3	UIU age	**SUPPORTED: 9.53-10.55 joint window**	P 10.46 | L 10.6 | R 10.55	The joint window from Gajah Hitam-1, Rusa Timur-1, Glayzer-3 and Solisip-1 is 9.53-10.55 Ma. Project 10.46 passes at all four wells. Legacy 10.6 fails at Glayzer-3, where NN9 is present below the pick. The recalibrated 10.55 sits on the edge, needing zero NN9 below the pick at Glayzer-3, where 50 m are observed.	Confirm Glayzer-3 NN9 (2060-2110 MD) is in situ, not caved.
C4	LIU age	**SUPPORTS PROJECT (1 well)**	P 11.63 | L 12.0 | R 11.90	Imbok-1 'Top Stage IVA (Uncon.)' at 1303 MD lies inside a 580 m NN7 span (1090-1670 MD): window 10.89-11.90 Ma, and interpolation gives 11.26 Ma. Project 11.63 passes, legacy 12.0 fails, and recalibrated 11.90 sits on the edge. With one well this is a single witness only.	Check the Imbok-1 ST-1 datum (shallow IVB/UIU picks conflict with NN10 and are excluded). Find a second NN6/NN7-bracketed LIU pick.
C5	Horizon III (Top IVD / Base IVE)	**HOLD**	P 8.11 | L 6.7	Both wells with picks (Gajah Hitam-1, Maligan-1) put Horizon III inside NN11 (5.59-8.29), so both models pass at zone level. Maligan-1 places it in NN11C-D (upper NN11), which favours the younger legacy 6.7 Ma. Also, 8.11 is not a GTS2012 zone boundary (base NN11 = 8.29); its source needs checking.	Calibrate NN11 subzones (A-D) with the PETRONAS 2024 chart and re-read Horizon III at all NN11 wells.
C6	Upper Stage IV definitions (IVE-	**HOLD - DEFINITIONAL**	Base IVF: P 3.1 vs L ~5.5 |	Offsets of 1.7-2.4 Ma are far too large to be	Decide which definition is

	IVG)		Base IVG: P 2.0 vs L 3.8	calibration. The project scheme (Phase I p8: IVE = NN16-NN11) stretches IVE to ~5 Myr and compresses IVF-IVG into the last 3.1 Myr; the legacy scheme keeps IVF = NN12-NN16 and IVG from 3.8 Ma. These are two different stage definitions under the same names.	canonical before Phase II mapping. Then rename one set (e.g. IVE-P vs IVE-L) in all data.
C7	Base Stage III / BMU	**HOLD**	P 23.13 (BMU) | L ~21 (TB2.1)	The project equates Top II with BMU at 23.13; the legacy chart puts Base III at TB2.1 (~21). Possibly two surfaces of one Early Miocene event family (22-20 Ma deformation).	Out of Phase I scope (NN8 and below deferred). Carry both.
C8	Composite unconformities on highs	**RULE ADOPTED**	single-name picks vs amalgamated surfaces	'SRU' is picked at top IVA (Deluar-1), base IVF (Ketam-1), base IVF/G (South Lokan-1), top IVC (Maligan-1) and base IVD (7 wells). 'DRU' is picked at base IVC at Tiga Papan-2 where NN5-NN8 is missing. On highs the names attach to whichever erosion surface is deepest.	Tag such picks as composite (e.g. DRU+LIU+UIU) and exclude them from age calibration of the single surface.
C9	MMU (DG) vs DRU (Sabah)	**HOLD**	RU/MMU ~16 Ma (end of spreading) vs DRU 14.2-15.4	Copilot merges 'DRU/MMU'. They are different physics in different domains: rift-end drowning on the lower plate versus collision and inversion in the upper plate. They may be linked (end of spreading as continental crust jams the trench) and near-coeval, but are not one surface. Tiga	Trace both on a dip keyline across the trough; look for onlap of trough fill onto DG drowned platforms vs DRU truncation inboard.

				Papan-2's '(MMU)' label shows the term has leaked inboard.	
C10	Kinarut fan stage	**UNKNOWN**	pen chart: Kinarut = Stage IVC (H136, NN10) | Copilot v1.1 (citing Atlas Ch.5): Kinarut in Stage IVE	The Atlas chapter was not available to this build. Fan names are package names, not chronostratigraphic units.	Read Atlas_Chapter 5 Sabah_v1.1; resolve before using fan names in GDE maps.
C11	Text QA in endorsed decks/tables	**FIX BEFORE IRS#1 (Oct 2026)**	-	(a) Phase I p8 calls the NN9 Stage IVc horizon a 'mid-Miocene' unconformity; NN9 is early Tortonian (Late Miocene). (b) p8 places thrust-belt initiation in 'Bunguran and Baram'; Bunguran is Natuna (Indonesia). (c) The unified table labels Stage I 'Late Oligocene (Priabonian)'; Priabonian is Late Eocene. (d) The table labels IVA (top 11.63) 'Late Miocene (Tortonian)'; IVA is Langhian-Serravallian.	Editorial correction.
C12	Well-level QC flags	**OPEN**	-	Kudat-1: '?Crocker Fm' at 2336-2475 MD carries NN9a nannofossils (formation call or caving issue). Maligan-1: ?NN10D above NN11D-C (reversal -> reworking). Imbok-1: shallow UIU/IVB picks above NN10. Rusa Timur-1: 'NN1' at 463-873 MD (truncated label).	Raw occurrence charts + abundance to separate in-situ vs reworked (C3 of Copilot v0.1).
C13	Inboard palaeobathymetry	**CORRECTION**	Copilot: inboard deepwater	Wells show both, but diachronously: Stage III is bathyal inboard	Plot EOD-vs-age for all 35+ biostrat

			shale until ~3 Ma; Stage IVA 'shallow-marine/deltaic'	(Deluar-1, Ketam-1, South Lokan-1); IVA is coastal on inversion highs but bathyal in the outer shelf (Rusa Timur-1, Gajah Hitam-1). The shelf edge then steps NW, reaching Rusa Timur at SRU time and Gajah Hitam in IVE. Shoaling is time-transgressive, not one horizontal facies switch.	wells to map shelf-edge trajectory.
C14	Copilot v1.1 rendering	**SUPERSEDED**	-	Undeclared non-linear time axis; onshore Crocker drawn as Neogene depocentre (it was the eroding source); diapirs coloured as hiatus; unsourced volcanics; 'Base NN4' at 15.5 Ma (GTS2012 base NN4 = 17.95); stage blocks mix top- and base-of-stage conventions (e.g. 9.53-11.63 labelled IVC-IVB).	-
C15	Cause of the post-16 Ma apparent-age signals	**HOLD - THREE MODELS REMAIN LIVE**	A older magmatism + younger reworking (VIABLE, MED) | B polyphase magmatism (PLAUSIBLE, LOW-MED) | C apparent youth without new magma (VIABLE, MED)	Layang-Layang-1 gives young apparent ages (6.5-7.8 Ma, 13.6 Ma) interleaved with old ones (19.6-22.9 Ma) and no depth progression. Every young K-Ar result is either a low-radiogenic separate (5.0-5.2% 40Ar*) or a bulk sample; none is a primary-mineral crystallisation age. All three models fit the current data. Models A and C explain young apparent ages without a new magma pulse; Model B explains them by true younger emplacement. Bulk	Promotion tests: (1) mineral-specific dating of screened primary igneous phases; (2) independent, well-tied cross-cutting seismic geometry.

				and fine-fraction K-Ar cannot separate them. No model is preferred.	
C16	Apparent age vs crystallisation / emplacement age	**RULE ADOPTED**	K-Ar apparent age | crystallisation age | emplacement age	Aggregate K-Ar on >2 µm fractions and bulk samples mixes primary and secondary mineral domains. A K-Ar apparent age is a mixture age until the dated phase is shown to be primary and closed.	No apparent age in this chart is labelled an emplacement age.
C17	Intrusion vs redeposited volcaniclastics	**HOLD - BOTH MAY OCCUR**	in-situ intrusion AND/OR redeposited volcaniclastics, determined body by body	Volcanogenic sediment deposition and reworking are supported; possible in-situ dyke or sill geometries are interpreted (both reported by the enterprise validation dossier from the LL-1 post-drill study; not verified in the working folder). The present evidence does not show that every volcanic-looking seismic body has the same origin. Rule: classify each body independently.	Per body: well-tied seismic geometry; contact petrography; chilled margin or baking; clast vs coherent groundmass texture; chemostratigraphic continuity; thermal alteration of host sediment; age relationship between body and host.
C18	Local proof vs regional analogue	**RULE ADOPTED**	local apparent ages | Mt Kinabalu granite 7.85-7.22 Ma | SCS spreading cessation 16-20 Ma (preliminary)	Regional Late Miocene magmatism (Mt Kinabalu) and post-spreading timing make a young local event plausible, but they are different domains and different plumbing. They are kept in a separate analogue ledger and never plotted as local evidence.	Only local, mineral-specific ages promote Phase C (see confidence ladder).

Phase C — candidate evidence (non-canonical)

verdict: HOLD for causal interpretation **canonical: false** **confidence LOW-MED**

INTERPRET constrained by DATA Local apparent-age evidence exists; emplacement age unresolved.

Warning. Aggregate K-Ar measurements include mixed primary and secondary mineral domains. Apparent ages must not be treated as emplacement ages without petrographic separation and mineral-specific dating.

Depth order. Depth order does not produce a simple age progression: 7.8, 6.5, 22.9, 7.3, 19.6, 13.6, 17.0 Ma from 4077 to 4293 m MD.

Three competing working models — none is preferred

Confidence ladder

▶ LOW-MED (current)

- local apparent-age signals younger than 16 Ma
- older 19.6–22.9 Ma apparent-age population
- regional younger-magmatism analogues
- ambiguous mineral domains and geometry

MED

EITHER concordant younger-than-16 Ma primary-mineral crystallisation ages from two screened in-situ samples
OR one well-tied intrusion demonstrably younger than a secure post-16 Ma horizon

MED-HIGH

BOTH independent tests above

Regional-analogue ledger (not local evidence)

Rock physics, mechanics and seismic relative timing

Lanes required before any Phase C body can be characterised. Nothing is filled in that the project data do not contain.

AVO / interface elastic properties

UNKNOWN: actual interface-average elastic properties not encoded

No AVO class (e.g. IIp, IV) is attached to a rock body. AVO is an interface response, not a lithology or age classifier.

Required per interface: exact interface, top or base, incident direction, polarity convention, wavelet phase, Vp above / below, Vs above / below,

Mechanical permissibility

UNKNOWN: no stress, pore-pressure or fault-property data encoded

Present-day stress is a permissibility constraint only. It never dates a Miocene intrusion.

Required for any conduit-reactivation statement: palaeostress and its uncertainty, effective stress, pore or magma pressure, fault

RHOB above / below, uncertainty range, interface thickness, measured vs modelled status, tuning risk, anisotropy caveat.

orientation, mechanical properties, timing uncertainty.

Seismic relative timing

UNKNOWN: no candidate intrusive feature is encoded with mapped seismic geometry

Relationship rules and implied inequalities are strictly enforced before any magmatic event is promoted.

Biostratigraphic discipline

• Long-ranging species are expressed as 'not older than', 'not younger than', 'and/or older' or 'and/or younger'.
• No long-ranging occurrence is turned into a zonal top or absolute age without the raw occurrence chart and assemblage support.
• Reworking, caving, barren samples, abundance and preservation controls are carried where available.
• Absence of evidence is not evidence of absence.

Encoding status: PARTIAL - the v2 well windows use zone assignments as reported in the penetration chart. Raw occurrence charts, abundance, preservation and barren-sample flags are NOT encoded, so every window is a zone-level constraint, not a species-level age.
QC flags: See well 'excluded' entries and contradiction C12 (Maligan-1 reversal, Rusa Timur-1 truncated label, Imbok-1 datum conflict, Kudat-1 '?Crocker' with NN9a).

Well age windows

Well	Domain	Surface	Pick (MD m)	Pick label	Zones above	Zones below	Window (Ma)	Interp.
GAJAH HITAM-1	INB (outer shelf)	**H3**	458	B.S IVE (HOR III)	445-651.6 NN11	445-651.6 NN11	5.59–8.29	-
GAJAH HITAM-1	INB (outer shelf)	**SRU**	1658	B.S IVD (S.R.U)	651.6-1767.8 NN10	651.6-1767.8 NN10	8.29–9.53	9.41
GAJAH HITAM-1	INB (outer shelf)	**UIU**	2406	B.S IVC (U.I.U)	2167-2590.7 NN9/NN8	2167-2590.7 NN9/NN8	9.53–10.89	-
TIGA PAPAN-2	INB	**SRU**	1696.5	B.S IVD (SRU)	1641.5-1705 NN10	1641.5-1705 NN10	8.29–9.53	9.36
MALIGAN-1	INB	**H3**	688	B.S IVE (within	670-790 NN11	670-790 NN11	5.59–8.29	-

				NN11D-NN11C)				
MALIGAN-1	INB	**SRU**	1144	TOP S.IVC SEAL/SRU (within NN10A lower)	1110-1180 NN10	1110-1180 NN10	8.29–9.53	9.38
SOLISIP-1	INB (N Sabah)	**SRU**	1083.51	B.S IVD (NN11 recorded to 1100.27 MD, below the pick)	877.78-1100.27 NN11	877.78-1100.27 NN11	5.59–8.29	-
SOLISIP-1	INB (N Sabah)	**UIU**	1883.57	B.S IVC	1499.24-1637 NN9	1885.71-2021.64 NN7/NN6	9.53–13.53	-
GLAYZER-3	INB	**UIU**	2010	Base S.IVC / U.I.U	1800-1990 NN11/NN10	2060-2110 NN9	5.59–10.55	-
RUSA TIMUR-1	INB	**SRU**	1674	B.S IVD (S.R.U)	873-1674 NN11/NN10	1674-1909 NN10/NN9	5.59–10.55	-
RUSA TIMUR-1	INB	**UIU**	1909	B.S IVC (U.I.U)	1674-1909 NN10/NN9	1909-2074 NN8	8.29–10.89	-
RUSA TIMUR-1	INB	**LIU**	2074	B.S IVB (L.I.U)	1909-2074 NN8		10.55–open	-
IMBOK-1	INB	**LIU**	1303	Top Stage IVA (Uncon.)	1090-1670 NN7	1090-1670 NN7	10.89–11.9	11.26

Phase C — candidate evidence (non-canonical)

verdict: HOLD for causal interpretation **canonical: false** **confidence LOW-MED**

INTERPRET constrained by DATA Local apparent-age evidence exists; emplacement age unresolved.

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QC flags: See well 'excluded' entries and contradiction C12 (Maligan-1 reversal, Rusa Timur-1 truncated label, Imbok-1 datum conflict, Kudat-1 '?Crocker' with NN9a).

Source ledger

ID	Kind	Source	Classification	Used for
SRC-PEN	DATA	Well Penetration Chart - Kinabalu Basin.xlsx (A-R MASTER PEN_CHART, Sheet1, pintasan-sukaugaya)	PETRONAS internal	Well NN zones, surface picks; l stage-base age stage mapping
SRC-UNI	DATA	Kinabalu_Basin_Unified_Stratigraphic_Chart_GEOX.xlsx (Key_Boundaries)	PETRONAS internal	Project marker 2.0 / IVE 3.1 / IV IVC 9.53 / IVB 1 11.63 / III 14.2 34.1.
SRC-P1	DATA	IBS Kinabalu Basin (Miocene Play) Phase I - Tectonostratigraphic Framework (TAC, May 2026)	PETRONAS internal	p8: Stage IVe = NN11, IVd = NN IVc = NN9; NN8 out of scope. p system mandat
SRC-NOM	DATA	KINABALU_BASIN_NOMENCLATURE_CORRECTIONS.xlsx	PETRONAS internal	NN label hygier legend (HIN 0-3 Lower Bathyal m).
SRC-MEMO	DATA	KL2_KINABALU_VALIDATION_MEMO.md (2026-09-07)	internal	KL2V2 roster re datum to Strati Chart of Malays V1.2024; veloci uncertainty +-2
SRC-IMG1	DATA	PETRONAS Chronostratigraphic Chart (2023) - KL8/KL5/KL1/KL3 transects	PETRONAS (Open)	Sabah Platform carbonates (Fal

				Merak, Cendera Setap/Melingan Belait, Liang, Temburong/Cro toe thrust, Saba sequence NP14
SRC-IMG2	DATA	South China Sea Events - TTE internal chart (PETRONAS-TotalEnergies; after Le Pourhiet et al. 2018, Pubellier et al. 2018)	Confidential - authorised for external distribution	ROU, IRU, BU, E RU/MMU; onset 30.8 Ma, ridge Ma, end spread Ages read from (+-0.5 Ma).
SRC-IMG3	DATA	Kinabalu Basin Geological Setting - Neogene stratigraphy IV-A to IV-G (SB412 slide)	Confidential - authorised for external distribution	DRU, LIU, UIU, S (North Sabah-P Wedge), Kudat Kalutan & Mant limestones; tec stages.
SRC-CP01	PRIOR	Copilot Kinabalu Earth-Event Chart v0.1 (json/csv/png)	internal	Audited - super
SRC-CP11	PRIOR	Copilot Kinabalu Visual Chronostratigraphic Chart v1.1 (png)	internal	Audited - see contradictions (
L-GTS2020	LIT	Gradstein et al. (2020) Geologic Time Scale 2020	public literature	Stage boundari
L-BACK12	LIT	Backman et al. (2012) Newsl. Stratigr. 45 - Neogene nannofossil zonation & GTS2012 calibration	public literature	NN base ages u
L-BKSA95	LIT	Berggren et al. (1995) SEPM SP54 - Cenozoic geochronology	public literature	Legacy calibrat NN5 15.6, base base NN11 8.6, NN16 3.83.
L-HAQ	LIT	Haq, Hardenbol & Vail (1987/88) - TB cycle chart	public literature	TB2.4 ~15.5, T ~10.5, TB3.4 ~ timescale).
L-LEV87	LIT	Levell (1987) GSM Bull. 21 - Sabah Stage IV subdivision	public literature	Stage IV A-G sc
L-HT93	LIT	Hazebroek & Tan (1993) - Tertiary tectonic evolution of NW Sabah continental margin	public literature	DRU, SRU, inboard/outboa
L-HUT05	LIT	Hutchison (2005) Geology of North-West Borneo	public literature	Sarawak Oroge Ma), Rajang/Cr Sabah Orogeny

L-HALL13	LIT	Hall (2013); Hall & Breitfeld (2017) - Borneo/SE Asia tectonics	public literature	Proto-SCS subd ended Early Mi crust underthru Sabah.
L-COT10	LIT	Cottam et al. (2010) J. Geol. Soc. 167 - Kinabalu granite U-Pb	public literature	Pluton emplace 7.22 Ma.
L-DG	LIT	Hinz & Schlueter (1985); Franke et al. (2008) - Dangerous Grounds rifting & carbonates	public literature	Rifted continen Oligo-Miocene drowning ~MM
L-LI14	LIT	Li et al. (2014) IODP Exp. 349 - SCS spreading history	public literature	Spreading ceas Ma.
L-DWFTB	LIT	Ingram et al. (2004); Morley et al. (2011); Hesse et al. (2009) - NW Borneo deepwater fold-thrust belt	public literature	Gravity-driven mobile shale, m Late Miocene-R
SRC-LL1-AGES	DATA	Layang-Layang-1 apparent-age table (6 K-Ar, 1 U-Pb lower intercept)	PETRONAS internal	Encoded exactl transcribed in A instruction 202 Enterprise prov reported AVAIL ENT-LL1 via SR DOSSIER); loca provenance PA provenance_ma
SRC-BIOREV	DATA	Sabah Kinabalu Basin biostratigraphic review (16.06.2026)	PETRONAS internal	Rule: long-rang expressed as 'r than' / 'not you 'and/or older' / younger'. Revie present in the v directory; rule a stated by ARIF.
SRC-ARIF-PHC	INT	ARIF Phase C framing (2026-10-12): candidate post-16 Ma magmatic or hydrothermal reactivation; Models A/B/C; confidence ladder	internal	Interpretive fra supplied by ARI observation.
SRC-ENT-DOSSIER	PRIOR	Enterprise validation dossier (M365 Copilot review, 2026-10-12)	PETRONAS internal	Reports that th table, the mixe caveat and the volcaniclastic a come from the sources below. verified in the v folder.
SRC-ENT-LL1	DATA	Layang-Layang-1 Post Drill Geological Study.pdf	PETRONAS internal (enterprise)	Reported prima for the six K-Ar fractions, 2σ, ra 40Ar and mixed

				caveat. Not in v folder; checksu page reference retrieved.
SRC-ENT-PKK-WCR	DATA	Pekaka-1 Well Completion Report.pdf	PETRONAS internal (enterprise)	Referenced by enterprise doss working folder.
SRC-ENT-PKK-BIO	DATA	APPENDIX 7 PEKAKA-1 BIOSTRATIGRAPHY REPORT.pdf	PETRONAS internal (enterprise)	Referenced by enterprise doss working folder.

Authority. GEOX computes and contrasts (local Python runtime; geox-direct MCP unavailable this session). ARIF / arifOS adjudicates.

Law. Surfaces are the stored objects; stages are derived intervals between surfaces. · Every age is model-scoped (PROJECT_2026, LEGACY_TB, LEGACY_RECAL). No silent averaging. · Biostratigraphy calibrates surfaces as age windows, it does not create surfaces. · No data is not nondeposition. Hiatus bodies are drawn only where wells or published charts show missing section. · Facies bodies are basin-scale synthesis, not measured thickness. Horizontal axis is domain position, not distance.

Convention. PROJECT_2026 ages (Unified GEOX table / Phase I) are TOP-of-stage ages; LEGACY_TB ages (A-R master penetration chart, Shell TB-cycle scheme) are BASE-of-stage ages. Both name the same physical surfaces: Top III = Base IVA = DRU; Top IVA = Base IVB = LIU; Top IVB = Base IVC = UIU; Top IVC = Base IVD = SRU; Top IVD = Base IVE = Horizon III; Top IVE = Base IVF = Horizon II; Top IVF = Base IVG = Horizon I.

v2.2 laws. Apparent K-Ar age is not automatically crystallisation, eruption or emplacement age. · Regional analogues increase plausibility but do not prove local plumbing continuity. · Present-day stress compatibility is a permissibility constraint, not age evidence. · AVO is an interface response, not a unique lithology or age classifier. · Never transform an open hypothesis into a canonical event merely to simplify rendering.

Classification. PETRONAS INTERNAL